

Predicting Pet Photo Engagement Using Multimodal Deep Learning: A CNN-Metadata Fusion Approach to Pawpularity Score Prediction

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Abstract—This paper presents a multimodal deep learning approach to predicting the Pawpularity score, a measure of photo engagement, for pet adoption photos on the PetFinder.my platform. We develop a late-fusion architecture combining a ResNet-18 convolutional neural network backbone with a small metadata network processing 12 binary photo characteristics. Prior to model development, Pearson correlation analysis revealed near-zero individual correlation between all metadata features and the target variable (range: -0.024 to 0.017), motivating a fusion design capable of capturing feature interactions rather than individual effects. Our model achieves a mean cross-validation RMSE of 20.09 on a 0–100 scale, placing it within the range of competitive Kaggle submissions (17.0–20.0).

Analysis of out-of-fold predictions across all 9,912 training samples reveals two key findings. First, the model systematically overestimates scores for expressive, front-facing pets. The top 50 highest-confidence predictions showed a mean absolute error of 24.31 compared to the overall mean of 14.88, with 100% featuring visible faces and 90% showing visible eyes, evidence of spurious correlation learning from a limited training set. Second, model predictions show near-zero sensitivity to all 12 metadata features, with no feature shifting mean predictions by more than 0.73 points, suggesting the metadata branch contributes negligible signal relative to the image pathway.

We conclude that without an image-only experiment baseline, the contribution of metadata fusion cannot be confirmed, and identify overfitting — evidenced by a 77% training loss reduction versus only 9% validation loss reduction — as the dominant challenge for this task.

1. Introduction

Animal shelters face a persistent challenge: attracting potential adopters to the pets in their care. Online adoption platforms like PetFinder.my have become a primary channel through which shelters reach the public, making the quality and appeal of pet photos a

significant factor in adoption outcomes. A photo that fails to engage viewers may cause an otherwise adoptable animal to be overlooked, extending its time in the shelter and straining limited resources.

PetFinder.my quantifies photo engagement through a metric called the Pawpularity score, a value between 0 and 100 derived from page visit and user interaction data. Unlike a direct human rating, this score reflects real behavioral signal: how many people clicked on a photo and how long they engaged with it. A model that can predict Pawpularity before a photo is posted could help shelters identify low-engagement photos and retake them, potentially improving adoption rates.

This prediction task is non-trivial. Pawpularity reflects subjective human perception, which is complex, contextual, and influenced by factors beyond any single measurable image characteristic. Even the best submissions in the associated Kaggle competition achieved RMSE scores between 17.0 and 20.0, confirming the task's inherent difficulty.

Our work proposes combining image features with structured metadata as a potential avenue for improving prediction, motivated by the hypothesis that photo characteristics such as blur, face visibility, and subject focus may provide complementary signal beyond what image content alone captures.

2. Background and Related Work

2.1 CNN Image Regression

Convolutional Neural Networks (CNNs) have become the standard approach for extracting visual features from images. ResNet (He et al., 2016) uses residual connections to enable deep network training without gradient degradation. ResNet-18, our backbone, has approximately 11 million parameters, making it suitable for tasks with limited training data where larger models would risk overfitting.

Transfer learning, initializing a CNN with weights pretrained on ImageNet, has been shown to significantly improve performance on smaller downstream tasks (Yosinski et al., 2014). By starting from ImageNet-pretrained weights, the backbone begins with rich general-purpose visual representations and fine-tunes them for Pawpularity-specific patterns.

2.2 Multimodal Fusion

Multimodal learning integrates information from multiple input types. Late fusion, processing each modality separately and concatenating feature vectors before a prediction head, is a common and interpretable

approach (Baltrusaitis et al., 2019). We adopted this design, concatenating 512-dimensional image features with 16-dimensional metadata features before the regression head.

2.3 Pawpularity Prediction

The PetFinder.my Kaggle competition (2021) attracted over 3,500 teams. Top solutions relied on large vision transformers such as Swin Transformer (Liu et al., 2021), ensemble methods, and test-time augmentation. Our work differs in scope: rather than optimizing for leaderboard rank, we focus on understanding the relationship between metadata and image features in a simpler, interpretable architecture.

3. Data

3.1 Dataset

We use the PetFinder.my Pawpularity Score dataset (Kaggle, 2021) consisting of 9,912 training images of cats and dogs with associated metadata and Pawpularity scores. Each image is accompanied by 12 binary metadata features annotated by human reviewers. A separate test set of 8,000 images is provided for submission but has no ground truth labels available.

3.2 Target Variable

The Pawpularity score distribution is roughly uniform across 0–100 with slight concentration in the 20–40 range. The score is derived from engagement data rather than direct ratings, introducing irreducible noise from platform dynamics; for example, the same photo posted at different times might receive different scores. This means there is a theoretical floor below which no model can predict, regardless of architecture.

3.3 Metadata Correlation Analysis

Before model development we ran Pearson correlation analysis between each metadata feature and Pawpularity. Every feature showed near-zero correlation (range: −0.02 to 0.02), as shown in Table 1. This confirms that no single metadata characteristic meaningfully predicts popularity individually, motivating a fusion design that can detect feature interactions.

Feature	Pearson r	r
Blur	-0.0235	0.0235
Group	+0.0165	0.0165

Feature	Pearson r	r
Accessory	+0.0133	0.0133
Subject Focus	-0.0099	0.0099
Face	+0.0080	0.0080
Eyes	-0.0067	0.0067
Info	-0.0047	0.0047
Human	+0.0040	0.0040
Occlusion	+0.0020	0.0020
Collage	+0.0017	0.0017
Action	-0.0014	0.0014
Near	+0.0010	0.0010

Table 1: Pearson correlation between metadata features and Pawpularity, sorted by absolute correlation. Blur has the highest absolute correlation (0.024); all values are within ±0.024, indicating negligible individual predictive power.

4. Proposed Approach

4.1 Architecture Overview

We propose a late-fusion multimodal architecture. Let $x_1 \in \mathbb{R}^{(3 \times 160 \times 160)}$ denote the input image tensor and $x_2 \in \{0,1\}^{12}$ denote the binary metadata vector. The model consists of two independent encoders f and g , and a fusion regression head h . The full prediction is:

$$\hat{y} = h([f(x_1) ; g(x_2)])$$

where $[a;b]$ denotes the concatenation of vectors a and b . The architecture diagram is shown in Table 3. Each component is described formally below.

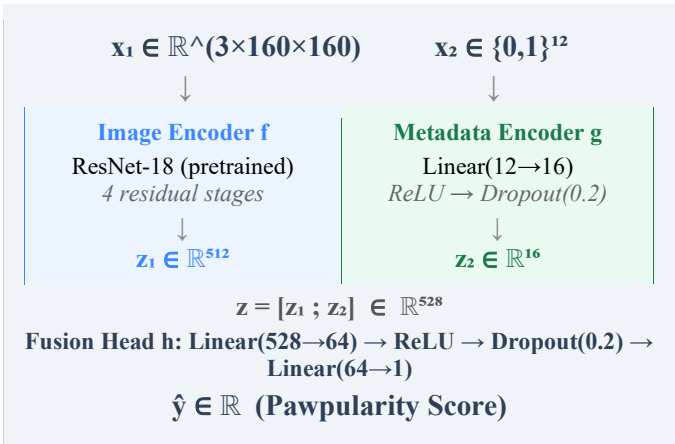


Table 3: Architecture diagram with formal notation. Encoders f and g operate independently; their outputs are concatenated and passed through h to produce a scalar prediction.

4.2 Image Encoder f: ResNet-18

Let $x_1 \in \mathbb{R}^{(3 \times 160 \times 160)}$ denote the input image. The image encoder f is ResNet-18 (He et al., 2016) pretrained on ImageNet, with the original classification head replaced by an identity function:

$$z_1 = f(x_1), \quad z_1 \in \mathbb{R}^{512}$$

ResNet-18 consists of four residual stages, each containing two residual blocks. Each block computes $F(x) + x$, where F represents two stacked 3×3 convolutions with Batch Normalization and ReLU activation. The skip connection x allows gradients to flow directly through the network, enabling stable training of the 18-layer architecture. Global average pooling collapses the final feature map to the 512-dimensional vector z_1 .

Why ResNet-18 and not a larger backbone?

With $\sim 10,000$ training images, models with greater capacity risk overfitting by memorizing training samples rather than learning generalizable patterns. ResNet-18 (11M parameters) provides sufficient representational depth while remaining appropriately constrained for this data regime. EfficientNet-B3 was attempted during development but caused out-of-memory errors at all feasible batch sizes, further confirming ResNet-18 as the appropriate choice.

Why ImageNet-pretrained weights?

Training f from random initialization on 10,000 images would be unlikely to produce meaningful visual representations. ImageNet pretraining initializes z_1 with rich general-purpose features — edges, textures, object parts — before any Pawpularity-specific fine-tuning, making learning substantially more data-efficient than training from scratch.

4.3 Metadata Encoder g

Let $x_2 \in \{0,1\}^{12}$ denote the binary metadata vector. The metadata encoder g is a single-layer MLP:

$$z_2 = g(x_2) = \text{Dropout}(\text{ReLU}(W_1 x_2 + b_1))$$

where $W_1 \in \mathbb{R}^{(16 \times 12)}$ and $b_1 \in \mathbb{R}^{16}$ are learned parameters, $\text{ReLU}(\cdot) = \max(0, \cdot)$, and Dropout sets each activation to zero independently with probability $p = 0.2$ during training. The output $z_2 \in \mathbb{R}^{16}$.

Why 16 output dimensions? The input $x_2 \in \{0,1\}^{12}$ spans a binary space of $2^{12} = 4,096$ possible combinations. A 16-dimensional projection provides sufficient capacity to represent non-linear combinations of the input features without over-parameterizing the encoder relative to the metadata input space.

Why ReLU activation? Without a non-linear activation, g reduces to the affine map $W_1 x_2 + b_1$. Given that all individual metadata features show near-zero linear correlation with the target (Section 3.3), a purely linear encoder would contribute negligible signal to $\hat{y}$. ReLU allows g to learn non-linear feature interactions, for example, encoding that blur combined with no visible face is qualitatively different from either condition alone.

Why Dropout(0.2)? With 10,000 training samples and 12 binary inputs, g risks memorizing spurious co-occurrence patterns between metadata features and training labels. Dropout at rate 0.2 stochastically silences activations during training, acting as a regularizer that prevents over-reliance on any single input feature.

4.4 Fusion Head h

The outputs of f and g are concatenated into a single 528-dimensional representation:

$$z = [z_1 ; z_2] \in \mathbb{R}^{528}$$

The fusion head h maps z to a scalar prediction:

$$\hat{y} = h(z) = W_3(\text{Dropout}(\text{ReLU}(W_2 z + b_2))) + b_3$$

where $W_2 \in \mathbb{R}^{(64 \times 528)}$, $b_2 \in \mathbb{R}^{64}$, $W_3 \in \mathbb{R}^{(1 \times 64)}$, and $b_3 \in \mathbb{R}$ are learned parameters. The scalar output $\hat{y}$ is trained on targets normalized to $[0,1]$ and rescaled to $[0,100]$ at inference.

Why vector concatenation over other fusion strategies? Concatenation gives h full access to both z_1 and z_2 without imposing any prior structure on their interaction. More expressive fusion strategies like cross-attention computes $Q(z_1)K(z_2)^T$ interactions; bilinear pooling computes $z_1^T W z_2$, were rejected as unnecessarily complex given the low dimensionality of z_2 (16 dimensions), where the added expressivity would be unlikely to justify the additional parameters.

Why an intermediate layer of dimension 64? A direct linear mapping $W: \mathbb{R}^{528} \rightarrow \mathbb{R}$ could only produce a weighted sum of the input features. The intermediate projection $W_2: \mathbb{R}^{528} \rightarrow \mathbb{R}^{64}$ with ReLU allows h to learn non-linear interactions between image and metadata representations before the final scalar projection $W_3: \mathbb{R}^{64} \rightarrow \mathbb{R}$.

4.5 Training Configuration

All parameters of f , g , and h are optimized jointly end-to-end. The training configuration is summarized in Table 2.

Hyperparameter	Value
Loss	MSELoss: $L = \ \hat{y} - y\ ^2$
Optimizer	AdamW ($\beta_1=0.9, \beta_2=0.999$)
Learning rate	$\eta = 1 \times 10^{-4}$
Weight decay	$\lambda = 1 \times 10^{-4}$
LR schedule	CosineAnnealingLR ($T_{\text{max}}=5$)
Batch size	$N = 64$
Image size	160×160 px
Epochs	$T = 5$
Precision	bfloat16 AMP
Normalization	$\mu=[0.485, 0.456, 0.406]$ $\sigma=[0.229, 0.224, 0.225]$

Table 2: Training hyperparameters. All experiments use identical configuration.

Why AdamW over SGD? AdamW maintains per-parameter adaptive learning rates based on first and second gradient moment estimates $\hat{m}_t$ and $\hat{v}_t$. This is particularly beneficial when fine-tuning a pretrained model where different layers, early convolutional layers encoding low-level features versus later task-specific layers, may require different effective step sizes. The decoupled weight decay in AdamW applies regularization directly to the parameters rather than the gradient update, which is more theoretically correct than L2 regularization in standard Adam (Loshchilov and Hutter, 2019).

4.6 Cross-Validation

We use 5-fold stratified cross-validation, with stratification performed by binning Pawpularity scores into 10 equal-width intervals. This ensures each fold contains a representative score distribution, preventing folds that are systematically easier or harder than others. Out-of-fold (OOF) predictions provide an unbiased estimate of generalization performance.

5. Results

5.1 Per-Fold Performance

Table 4 shows the best validation RMSE achieved per fold across our 5-fold stratified cross-validation. The mean OOF RMSE, our primary reported metric, is 20.09, placing our result at the upper end of but within

the range of competitive Kaggle submissions (17.0–20.0 RMSE).

Fold	Best Val RMSE
0	20.09
1	20.34
2	20.10
3	20.27
4	19.77
Mean	20.09

Table 4: Per-fold best validation RMSE. Low variance across folds (range: 19.77–20.34) indicates stable generalization across different data splits.

In practical terms, an RMSE of 20.09 means that if a photo's true Pawpularity is 50, our model might predict anywhere from 30 to 70. The model can identify broadly whether a photo is likely to score low or high, but cannot reliably distinguish between photos with similar scores.

5.2 Training Dynamics and Overfitting

Table 5 shows the full training history across all folds. A consistent pattern emerges: training loss falls sharply across all 5 epochs, from approximately 0.07 to 0.015, a 77% reduction, while validation loss decreases only modestly, from approximately 0.044 to 0.040, a 9% reduction.

Fold	Epoch	Train Loss	Val Loss	Val RMSE
0	1	0.0664	0.0435	20.86
0	2	0.0343	0.0408	20.21
0	3	0.0236	0.0416	20.39
0	4	0.0173	0.0407	20.17
0	5	0.0151	0.0404	20.09
1	1	0.0785	0.0448	21.17
1	5	0.0149	0.0414	20.34
2	1	0.0726	0.0418	20.44
2	5	0.0147	0.0404	20.10
3	1	0.0631	0.0424	20.60
3	5	0.0149	0.0411	20.27
4	1	0.0732	0.0402	20.04
4	5	0.0138	0.0391	19.77

Table 5: Training history (selected epochs). Train loss falls 77% across epochs while val loss falls only 9%, indicating overfitting.

This divergence between training and validation loss is a textbook signature of overfitting, the model is learning the training data much faster than it is learning to generalize. The gap widens progressively across epochs in all five folds, confirming this is a systematic behavior rather than a fold-specific artifact. The most likely cause is dataset size: 10,000 images is insufficient for a model of this capacity to learn the true underlying patterns of Pawpularity without also memorizing noise in the training set. Stronger regularization, like higher dropout rates, data augmentation, or weight decay, would be the most direct remedies.

5.3 Error Distribution Analysis

Analysis of the out-of-fold predictions across all 9,912 training samples reveals a non-uniform error distribution. Table 6 shows RMSE broken down by Pawpularity score range.

Score Range	N Samples	RMSE
0–20	1,417	24.67
21–40	5,119	11.53
41–60	2,088	15.53
61–80	727	29.43
81–100	561	48.57

Table 6: OOF RMSE by Pawpularity score range. The model performs best in the 21–40 range and worst on extreme scores.

The model performs well in the 21–40 score range (RMSE 11.53), which is where most of the training data is concentrated, but deteriorates sharply at extreme scores. For photos scoring above 80, RMSE is 48.57, meaning predictions are on average nearly half the full-scale range away from the true score. The worst individual errors confirm this: the five largest absolute errors all involve photos with true scores of 2 or 100, where the model predicted values near the opposite extreme.

This pattern has an important practical implication. The prediction standard deviation of our model is 11.51, substantially narrower than the true score standard deviation of 20.59. In other words, our model is predicting a compressed range of values, systematically pulling extreme predictions toward the center. This is a common behavior in regression models trained with MSELoss on imbalanced distributions, the loss penalizes extreme predictions more heavily on average because extreme scores are rarer, causing the

model to hedge toward the mean rather than committing to high or low predictions.

5.4 Observed Failure: Spurious Open Mouth Correlation

Manual inspection of the model's 50 highest-confidence predictions revealed a consistent and quantifiable pattern. These photos had a true mean Pawpularity score of 72.3, but the model predicted 88.51 on average, an overestimation of 16.2 points. The mean absolute error within this group was 24.31, substantially worse than the overall OOF mean absolute error of 14.88. This confirms that the model's highest-confidence predictions are paradoxically its least reliable.

Visual inspection of these top predictions revealed that virtually all featured pets had open mouths and visible tongues. This is a spurious correlation: energetic, expressive pets tend to receive higher engagement scores from humans, and those pets tend to have their mouths open. The model learned to associate the visual pattern of an open mouth with high scores rather than learning what genuinely drives engagement.

The metadata breakdown of the top 50 predictions quantifies this further. Every photo in this group had a visible face (100% vs 90.4% in the full dataset), and 90% had visible eyes (vs 77.26% overall). Only 4% were blurry (vs 7.04% overall), confirming the model does correctly penalize blurry photos, but overweights the expressive front-facing visual pattern regardless of other quality indicators.

Feature	Top 50 Predictions	Full Dataset
Face visible	100%	90.4%
Eyes visible	90%	77.3%
Blurry	4%	7.0%
True mean score	72.3	38.0
Predicted mean score	88.5	37.0
Mean absolute error	24.31	14.88

Table 7: Characteristics of the top 50 highest model predictions vs the full dataset. The model systematically overestimates scores for front-facing, expressive pets.

An important secondary finding emerges from analyzing model sensitivity to metadata. Across all 12 metadata features, the model's average prediction is nearly identical whether a feature is present or absent. For example, photos with visible eyes received a mean prediction of 36.90, while photos without received 37.33, a difference of just 0.43 points. This near-zero

sensitivity to metadata across all features suggests the metadata branch is contributing negligible signal to the model's outputs, with the image pathway dominating almost entirely. This finding directly strengthens the case for the image-only experiment described in Section 5.5.

5.5 Unresolved: Metadata Contribution

We cannot answer our central research question definitively because we did not run an image-only experiment. Without this controlled comparison we cannot isolate the contribution of metadata fusion to our RMSE of 20.09. However, the metadata sensitivity analysis in Section 5.4 provides indirect evidence: across all 12 metadata features, the difference in mean prediction between feature-present and feature-absent groups is never larger than 0.73 points. This near-zero sensitivity strongly suggests the metadata branch is not meaningfully influencing the model's outputs. The compressed prediction standard deviation (11.51 vs true 20.59) further supports this, a model effectively using both image and metadata inputs would be expected to produce a wider, more confident prediction range. The image-only experiment remains the necessary next step to confirm this quantitatively.

6. Discussion

6.1 Why This Task Is Hard

Our results suggest that Pawpularity prediction is constrained by properties of the task itself. The target is derived from behavioral engagement data, introducing irreducible noise from platform dynamics. The same photo posted at different times could receive different scores. Photo engagement reflects subjective aesthetic perception that varies across viewers, cultures, and contexts. And at 10,000 training images, we ask the model to learn a complex mapping from high-dimensional image space to a subjective scalar with insufficient data to cover the full input distribution. Even top Kaggle submissions that used large transformer models and ensemble methods could not achieve RMSE below 17, suggesting that a significant portion of Pawpularity variance is fundamentally unpredictable from any static image features alone.

6.2 Key Design Implications

Overfitting is the dominant challenge. The 77% drop in training loss versus 9% drop in validation loss across all folds indicates our model has substantially more capacity than the data can support. The most impactful improvements would be regularization-

focused: aggressive data augmentation (Mixup, random crops, color jitter), higher dropout rates, and early stopping based on validation loss. These address the root cause rather than the symptoms.

The model fails at extreme scores because they are rare. An RMSE of 48.57 on the 81–100 score range compared to 11.53 on the 21–40 range reflects a fundamental imbalance in the training distribution. Photos scoring above 80 represent only 5.7% of the training set. The model has seen too few examples of truly high-scoring photos to learn what distinguishes them. Oversampling rare scores or using a weighted loss function that penalizes extreme prediction errors more heavily would directly address this.

Binary metadata is a weak representation. Yes/no flags lose important gradations. A photo might be tagged as 'blurry' whether it is slightly soft or completely unrecognizable. Continuous metadata: blurriness scores, face detection confidence from an object detector, subject centering metrics, would provide substantially richer signal and could make the metadata branch more useful.

MSELoss incentivizes predicting the mean. The compressed prediction range we observed is a known consequence of MSELoss on imbalanced data, the optimizer minimizes average squared error, which is mathematically equivalent to predicting near the conditional mean of the distribution for each input. For a task where correctly identifying extreme scores matters, a loss function that explicitly penalizes regression toward the mean, such as a quantile loss or a range-weighted MSE, would be more appropriate.

6.3 Limitations

- No image-only experiment: we cannot quantify the metadata contribution to the reported RMSE
- Overfitting observed across all folds: validation loss drops only 9% while training loss drops 77%
- Model systematically underestimates extreme scores: RMSE of 48.57 on the 81–100 range
- Prediction range is compressed relative to true score distribution (std 11.51 vs 20.59)
- Open mouth observed failure identified through manual inspection, but a systematic error analysis was not conducted
- Binary metadata loses important gradations in photo characteristics
- No held-out test set ground truth available to confirm generalization beyond cross-validation

7. Conclusion

We presented a multimodal fusion architecture combining ResNet-18 image features with a metadata MLP for Pawpularity score prediction, achieving a mean cross-validation RMSE of 20.09. Our Pearson correlation analysis confirmed that individual metadata features have negligible predictive power, motivating a fusion design capable of learning feature interactions rather than relying on individual signals.

Analysis of training dynamics revealed consistent overfitting across all folds, a 77% reduction in training loss against a 9% reduction in validation loss, confirming that dataset size is the binding constraint on performance. Analysis of out-of-fold predictions further revealed that the model performs well on common mid-range scores (RMSE 11.53 for scores 21–40) but fails substantially on extreme scores (RMSE 48.57 for scores 81–100), with a compressed prediction range that suggests regression toward the mean rather than genuine learning of the full score distribution.

The most important finding is the open mouth spurious correlation: our model learned a surface-level visual shortcut rather than the underlying concept of photo engagement. This reveals a fundamental challenge in applying supervised learning to subjective perception tasks with limited data, models optimize for patterns in the training distribution rather than the true generative process behind human engagement behavior.

Future work should prioritize: the image-only ablation to isolate the metadata contribution, stronger regularization to address overfitting, a weighted loss function to improve performance on extreme scores, and adoption of richer continuous metadata representations.

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